

Comparative Analysis of Engineering Materials Using Microsoft Excel

Objective

To compare the mechanical and physical properties of common engineering materials using Microsoft Excel and derive insights for preliminary material selection.

Key Observations

- Titanium Grade 5 exhibits an excellent strength-to-weight ratio.
- Inconel 718 has the highest tensile strength in the dataset.
- Aluminium alloys provide low density for lightweight applications.
- Tool steels show the highest hardness values.
- Material selection depends on balancing strength, weight, hardness, and service conditions.

Conclusion

The analysis demonstrates how Excel can be used to compare engineering materials and support informed material selection using charts, summary statistics, and dashboards.